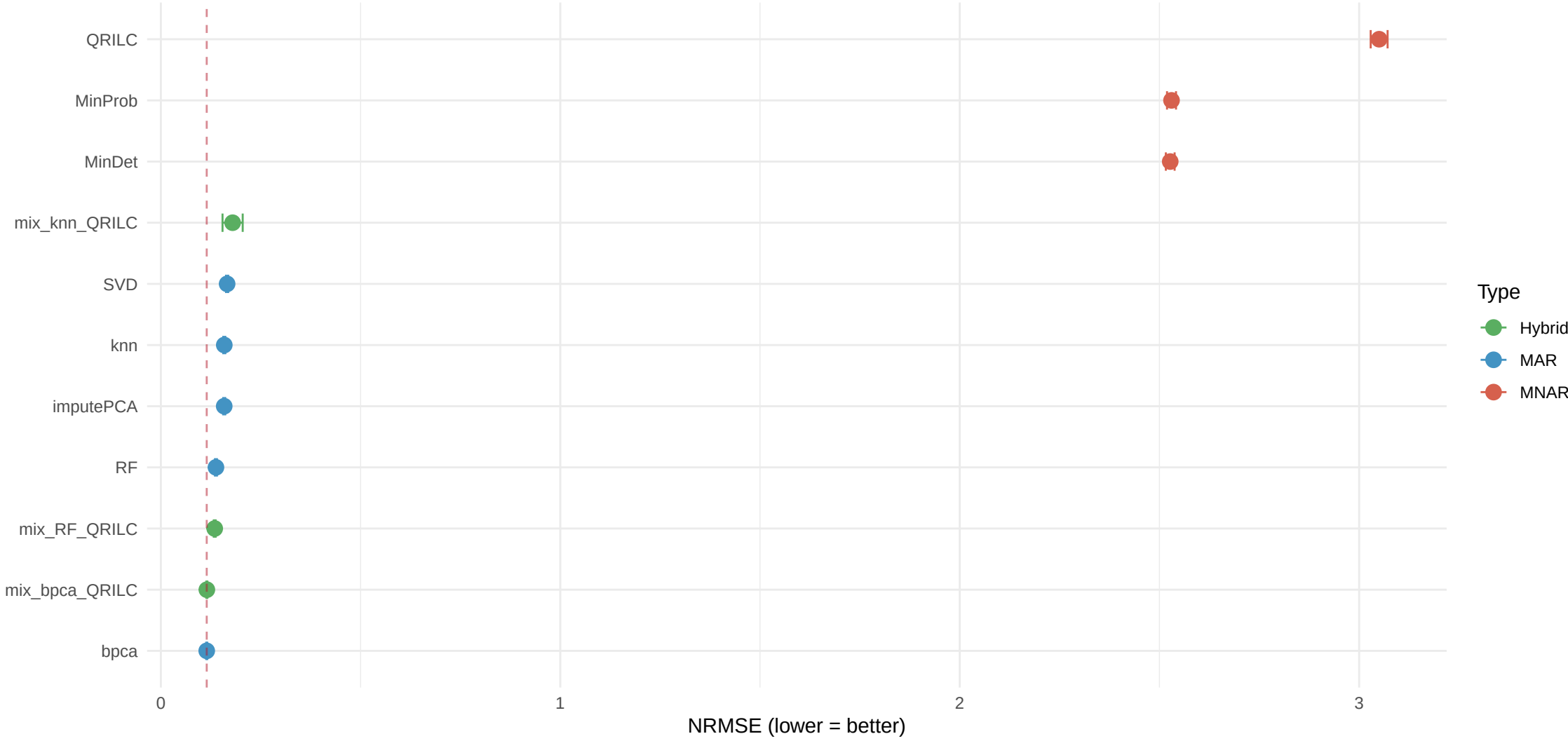
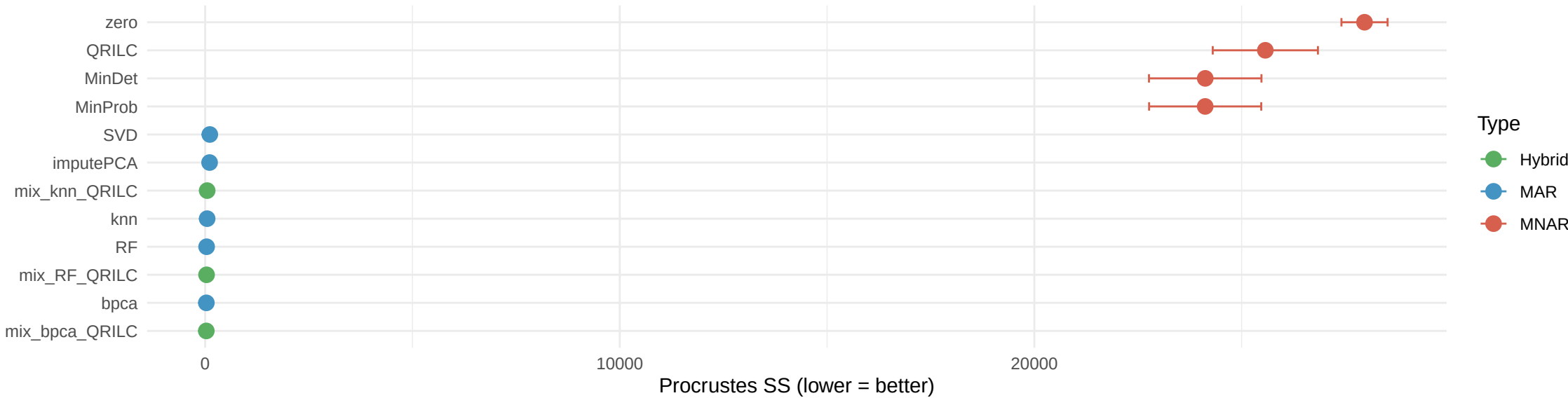


Imputation Benchmark

Imputation Benchmark
20 iter x 10% masked | Best: bpca (0.1146) | Off-scale: zero

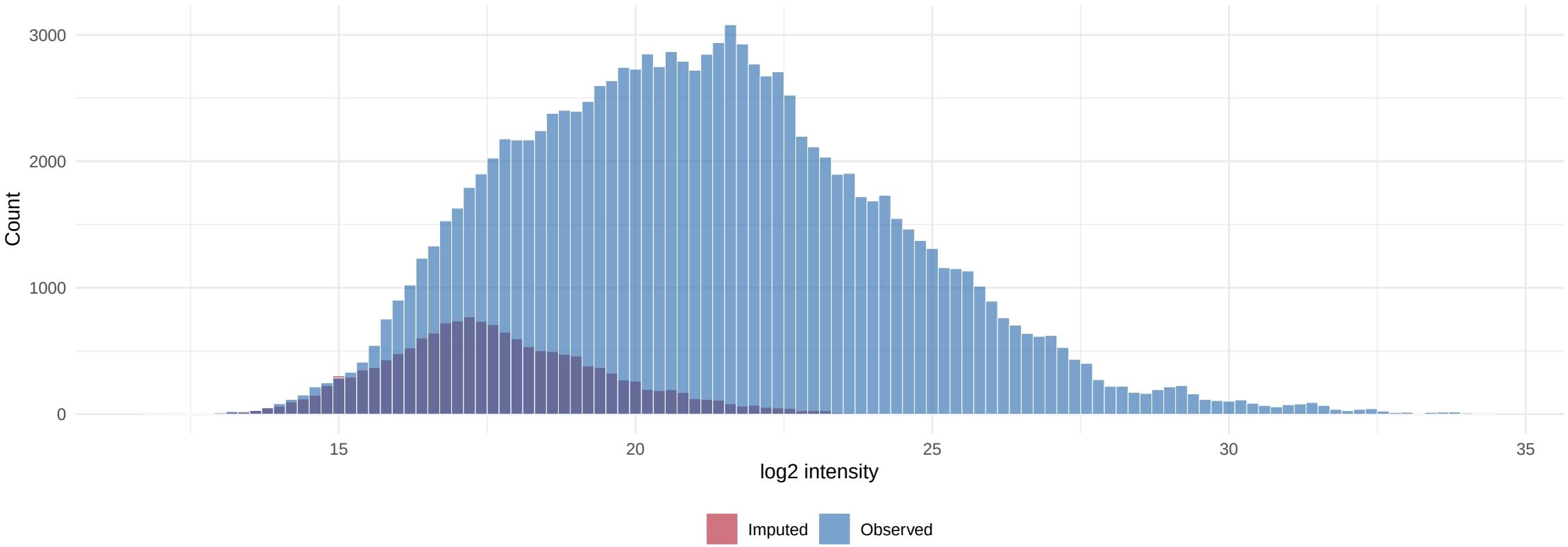


PCA Structure Preservation
Lower PSS = less geometric distortion of sample relationships

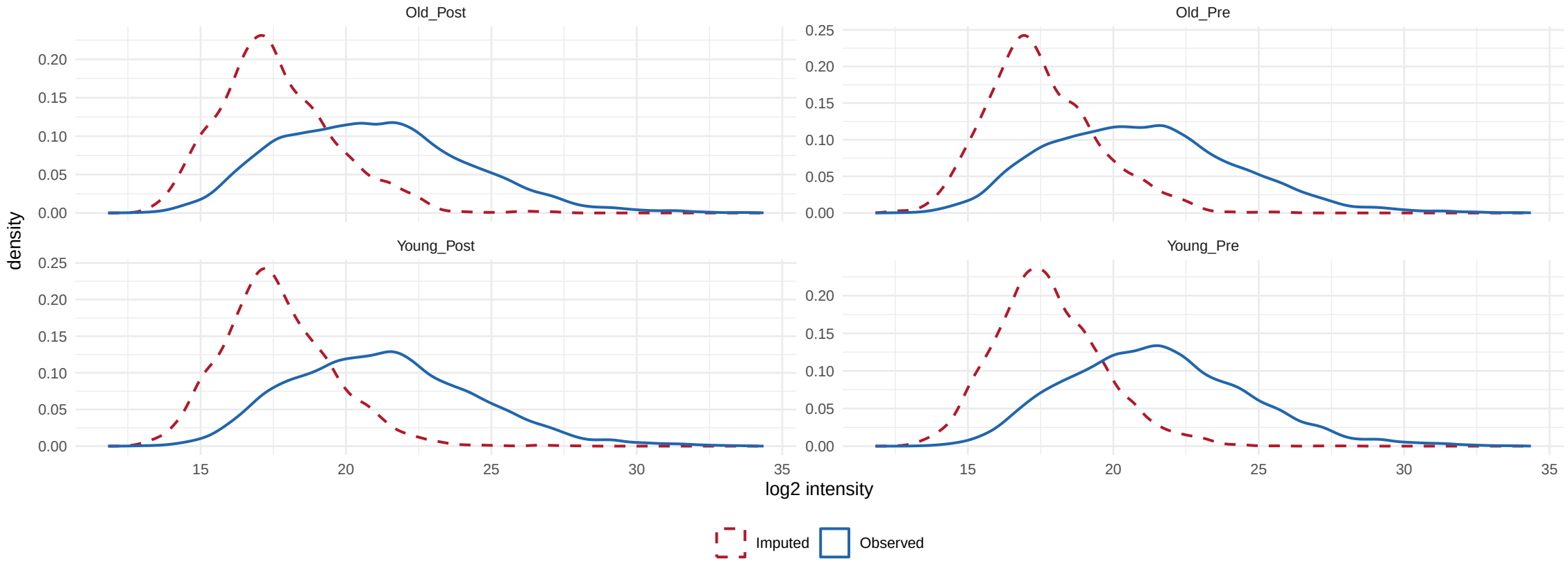


Post-Imputation Quality (bpca)
2124 proteins | 15389 values imputed (11.7%)

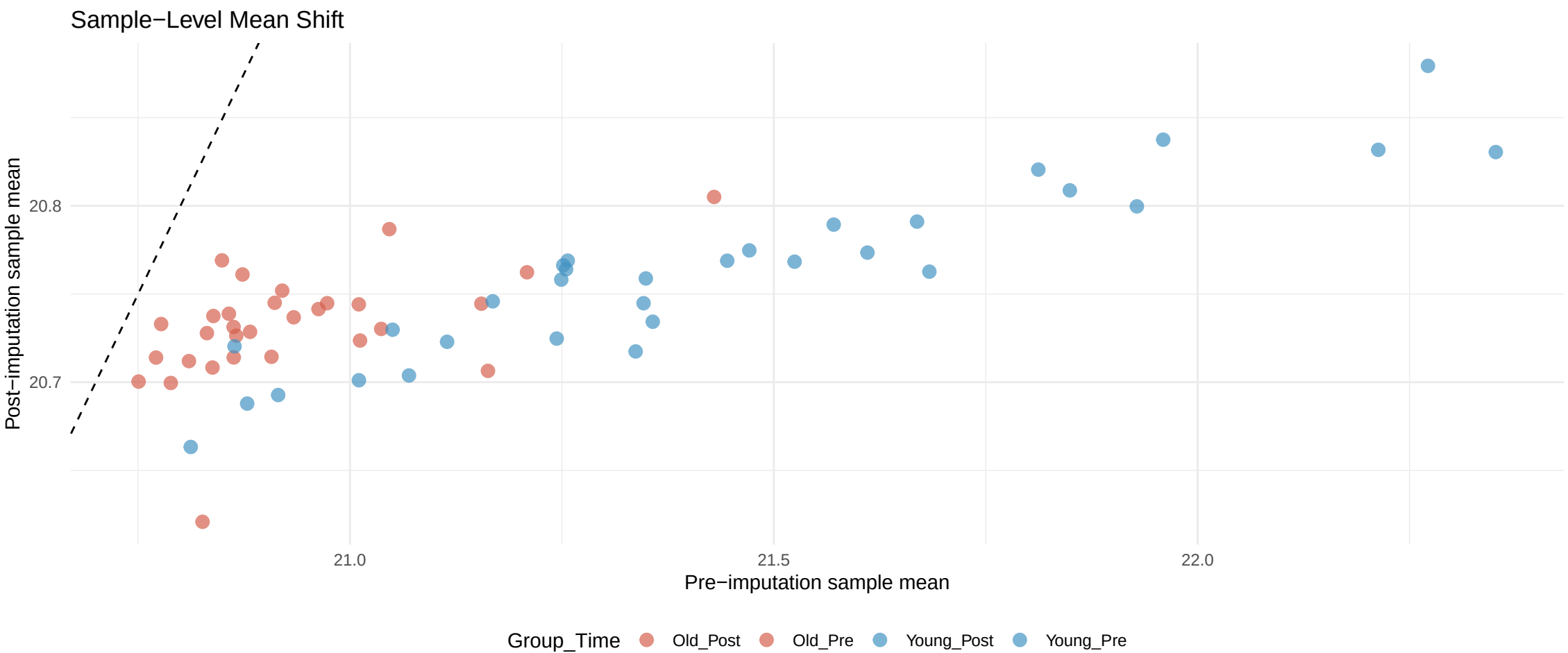
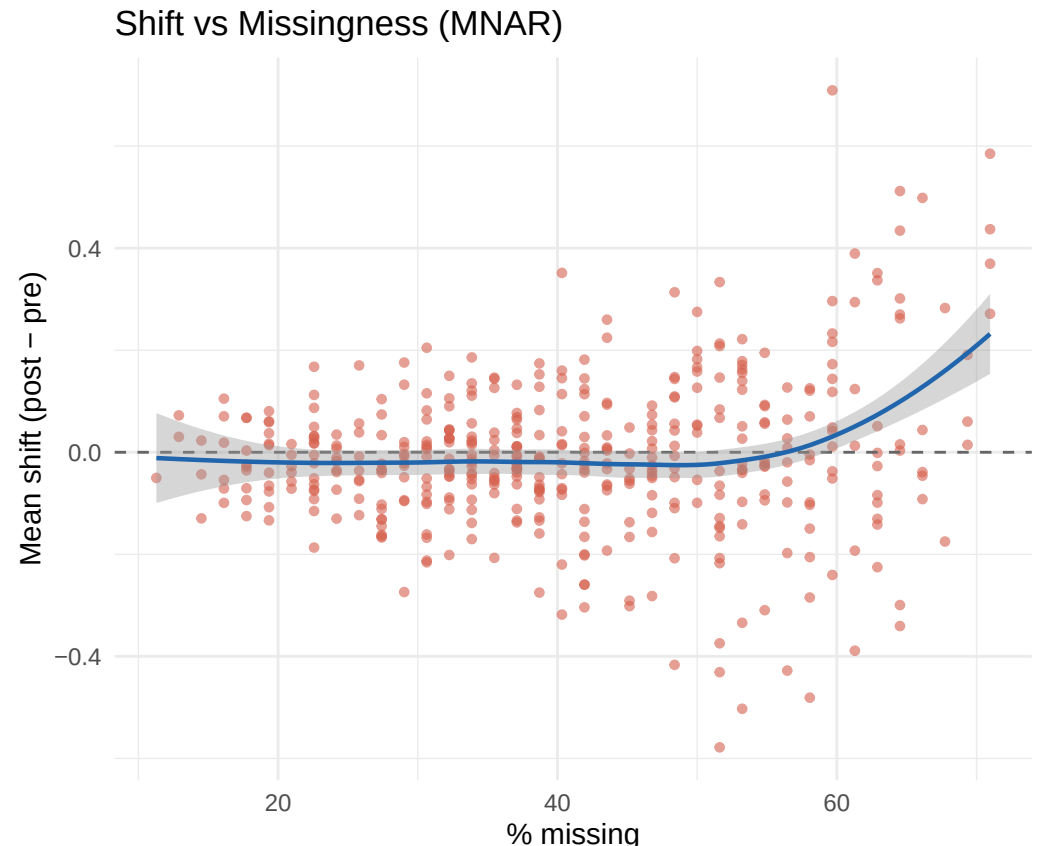
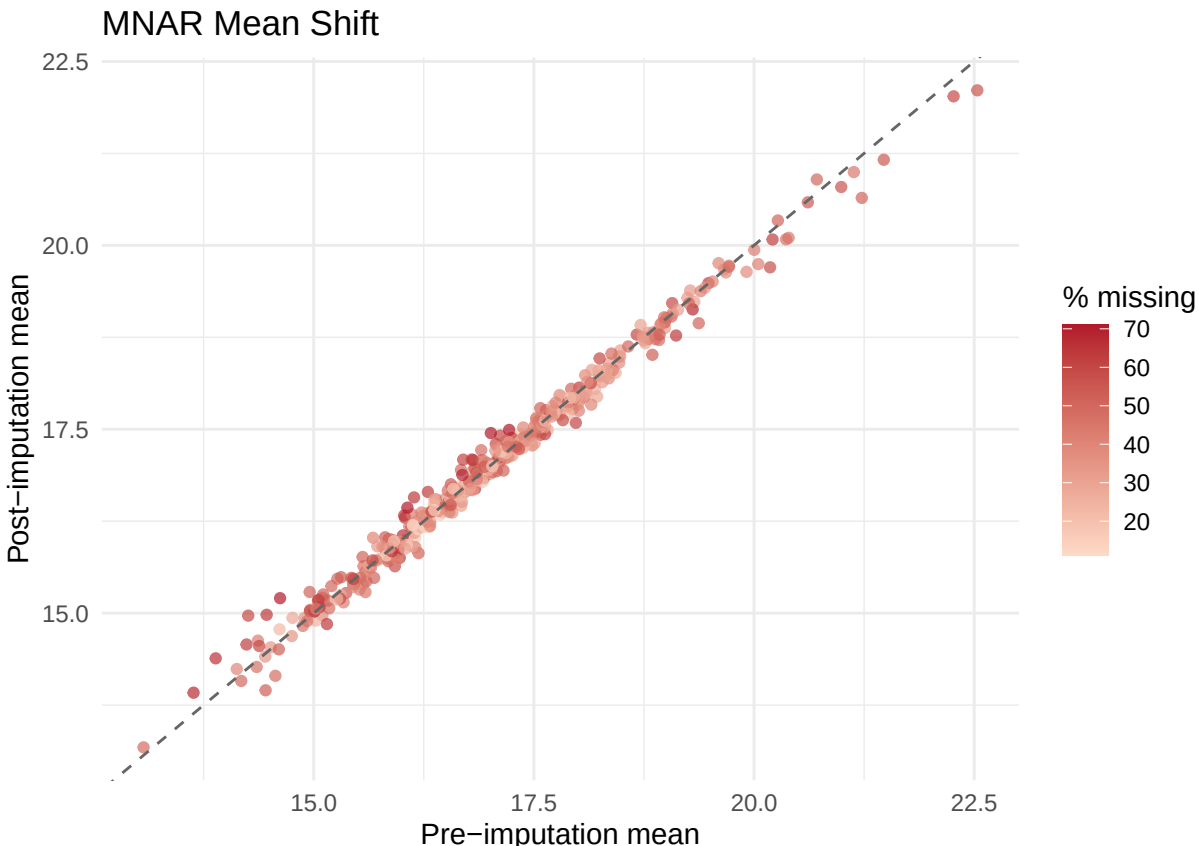
Observed vs Imputed Distributions
116,299 observed | 15,389 imputed



Observed vs Imputed by Group

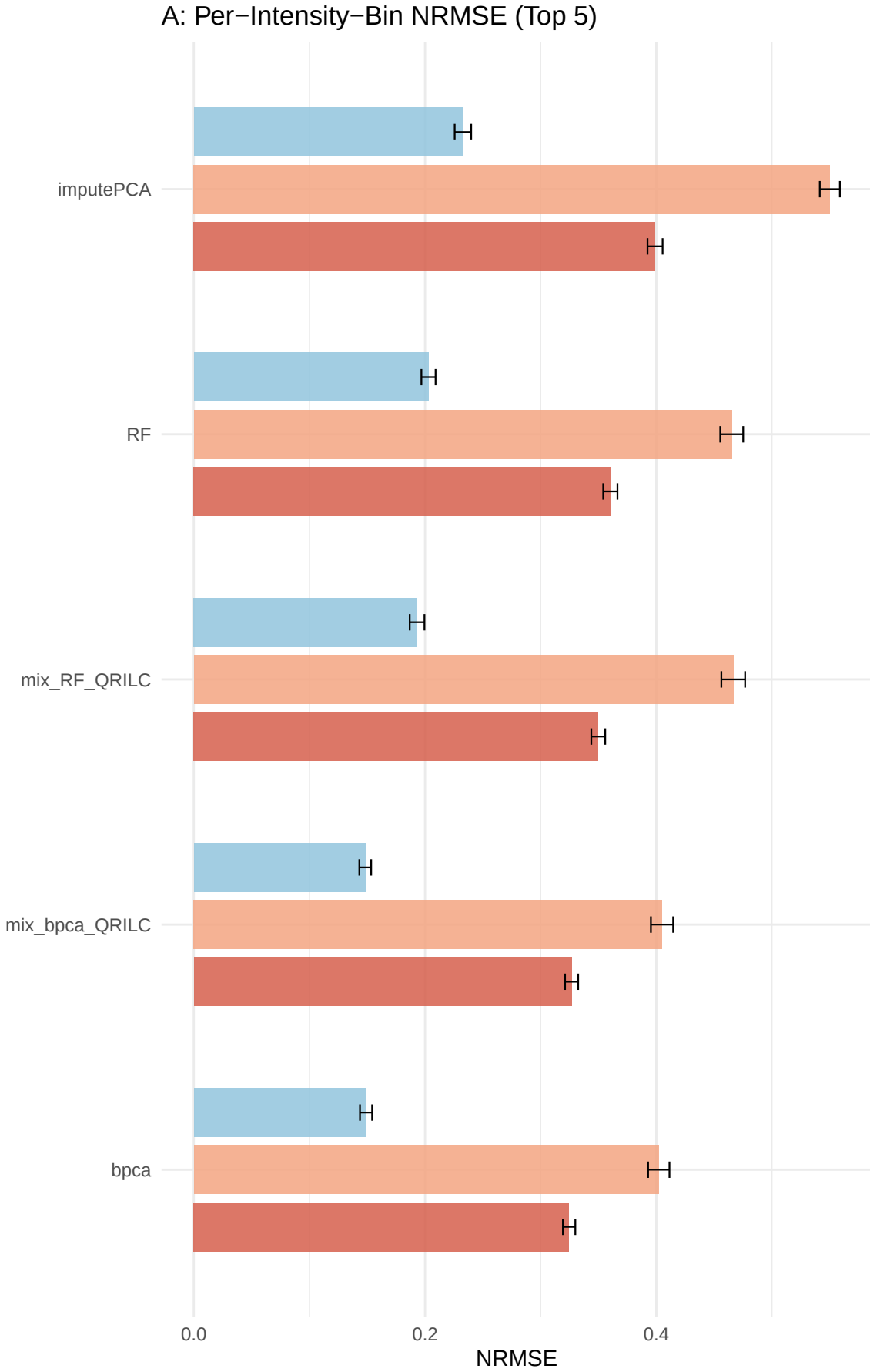


MNAR Audit (435 proteins) & Sample Shift



Extended Diagnostics (Top 5 Methods)

Per-intensity accuracy and method summary



B: Extended Summary

Intensity tertile	method	nrmse	pss	nrmse_high	nrmse_low	nrmse_mid
	bpca	0.1146	28.6447	0.1490	0.3246	0.4021
	mix_bpca_QRILC	0.1150	27.4107	0.1483	0.3267	0.4049
	mix_RF_QRILC	0.1343	33.5741	0.1931	0.3498	0.4665
	RF	0.1378	35.0977	0.2030	0.3602	0.4652
	imputePCA	0.1580	106.8614	0.2328	0.3989	0.5500

Per-Sample Imputation Accuracy

Per-Sample LOOCV (bpca, 20 iter × 10% mask)

High/Low mean NRMSE ratio: 1.57 | Ref line: bpca benchmark (0.1146)

